

EQUIPMENT CONTROLLER

Using MATLAB-Based GUI

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In this project, a MATLAB platform to control up to four electrical equipment is presented. Controlling electrical equipment from a remote location using a control panel is employed in many applications. An important application is in a process plant where an operator in a control room, sitting in front of a computer terminal and running a human-machine interface (HMI) application program for process monitoring and controlling, can effectively send hard, real-time control commands/signals to multiple actuators present on the field. A graphical user interface (GUI)-based HMI application enhances the user-friendliness of such an approach.

The authors' prototype and the MATLAB-based GUI front panel are shown in Figs 1 and 2, respectively.

Circuit and working

Components used for this project are described below.

Arduino Uno board. Arduino Uno is an AVR ATmega328P microcontroller (MCU)-based development board with six analogue input pins and 14 digital I/O pins. The MCU has 32kB ISP flash memory, 2kB RAM and 1kB EEPROM. The board provides the capability of serial communication via UART, SPI and I2C. The MCU can operate at a clock frequency of 16MHz. In this project, digital I/O pins 9, 10, 11 and 12 of Arduino are configured as output pins.

SPDT relay. A relay is an electromechanical switch. Relays used here are of single-pole, double-throw (SPDT) type. In this type of relay, output side has three terminals: normally-open (NO), normally-closed (NC) and common (C).

When the relays are not energised, NC and C terminals are short-circuited, and NO and C terminals are open-circuited. When relay coil is energised, NC and C terminals are

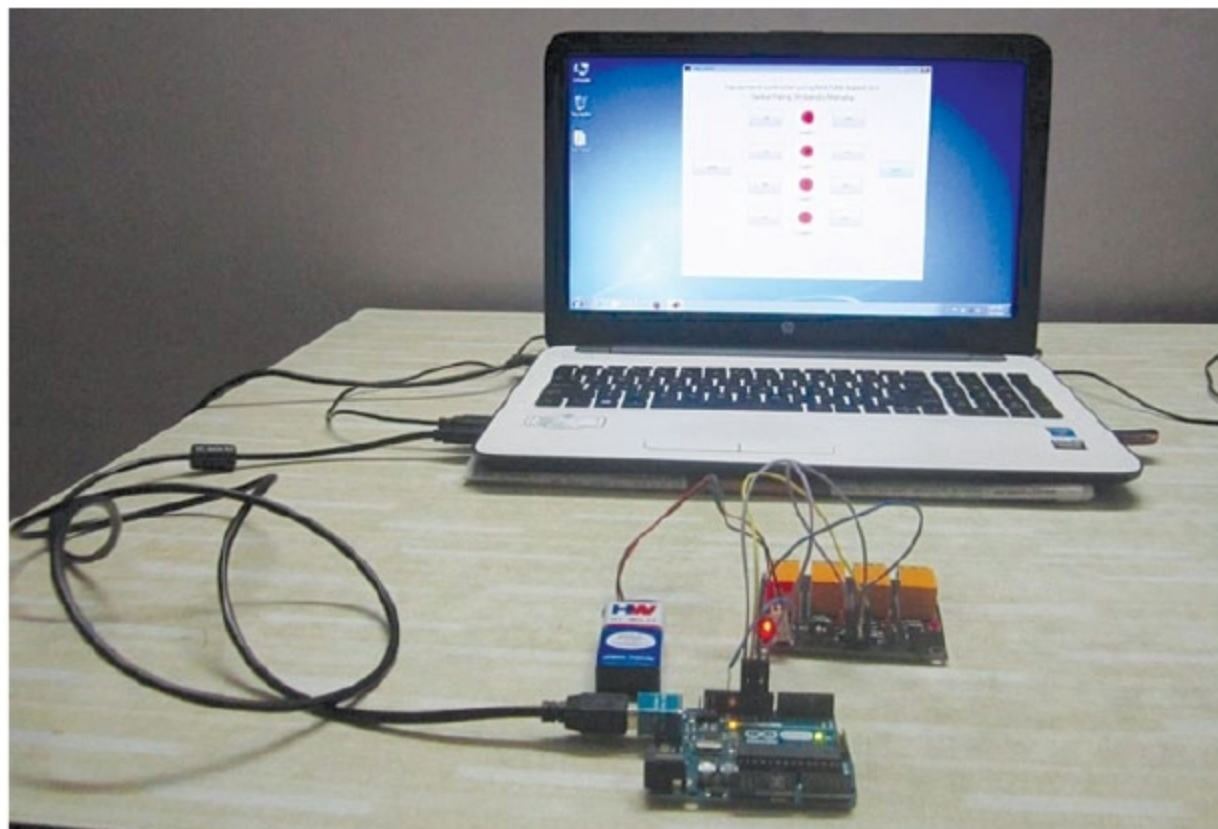


Fig. 1: Prototype of the equipment controller

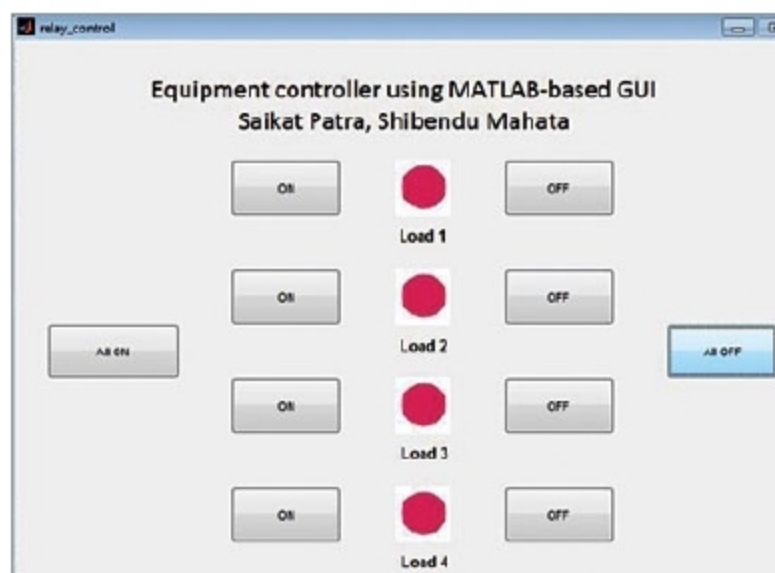


Fig. 2: MATLAB-based GUI

open-circuited, and NO and C terminals are short-circuited. The prototype of the relay board is shown in Fig. 3.

Block and circuit diagrams for controlling the relays are shown in Figs 4 and 5, respectively.

In this project, when you click on a particular pushbutton in the GUI, MATLAB program executes a callback function corresponding to that pushbutton in `relay_control.m`

MATLAB program. Within that function, instructions to set/reset digital I/O pins 9, 10, 11 and 12 of Arduino are executed to drive the relay as desired.

When a particular digital I/O pin is set high by clicking on the corresponding pushbutton using a mouse, the corresponding relay is energised. Four BC547 bipolar junction transistors (T1-T4) are additionally used as drivers because digital pins of

Arduino cannot directly source sufficient current to energise the relays.

Diodes D1 through D4 are used as protection diodes to prevent the sinking of current into Arduino digital pins. D5 through D8 are fly-back diodes that are used to prevent damage to electronic components due to back electromagnetic force (EMF) produced by inductive action of the coils of the relays.